

**Geodesics of Care:
A Scale-Free Geometry of Concern**

PIATRA . INSTITUTE

June 2026

Abstract

Care is usually studied as an emotion, a virtue, or a social practice. This paper develops a different primitive. A system cares for another, in the weak sense, when the second system's viability enters the first system's perception, model, valuation, and choice of action; it cares in the strong sense when that coupling is conscious, valued, and answerable to the other's consent and standing. The proposal is that this single relation, directed viability-coupling, runs from bacterial coordination through immune tolerance, animal attachment, human compassion, institutions, and artificial agents, and that its structure is geometric. We define a care-distance as the least-cost admissible path by which one agent comes to detect, represent, value, and improve another's viability without collapsing the other's agency, and we show that this distance is a directed dissimilarity: it is asymmetric, and it can violate the triangle inequality, so the moral world is not a circle around the self but a curved and broken manifold of concern. The vocabulary of that manifold, radius, diameter, depth, curvature, topology, and singularity, is given precise definitions and matched to established findings in moral psychology, which become measurements of curvature rather than a list of separate biases. A minimal control model, solved exactly, makes three structural features concrete: care-distance is a directed dissimilarity, asymmetric and triangle-violating; preserving the other's option-entropy has a measurable price in the other's expected viability, and that price is small while the welfare lost to coercion's failure is large; and the policy deformation required to help scales steeply with relational distance, so kin are an order of magnitude cheaper to reach than distant strangers. The model is solved exactly, so it shows that these structural features are realizable and internally coherent, not that any empirical agent exhibits these magnitudes. The account is organizational and formal rather than a new ethics, and it is bounded: rendering concern as navigable geometry measures coupling, cost, and the preservation of agency, and it does not measure, or explain, what concern feels like from inside.

1. Care Without Sentiment

The standard objects of a theory of care are feelings and the actions feelings prompt. Empathy is the sharing of another's state, compassion the wish to relieve it, and the moral life is then a matter of having the right feelings widely and reliably enough. This starting point makes two things hard to see. It makes the care delivered by a hospital or a welfare office, which feels almost nothing for any particular person, look like a degenerate case rather than a central one. And it makes the failures of care look like failures of feeling, to be repaired by exhortation, when many of them are structural and survive any amount of goodwill.

A more tractable primitive is available. Treat care as a relation between the regulatory life of one system and the viability of another. System A cares for system B , in the weak and purely functional sense, when variables that track B 's viability enter A 's perception, internal model, valuation, and selection of action, so that what happens to B changes what A does. This says nothing yet about feeling or obligation. It is the relation a thermostat would have to a second thermostat whose setpoint it had begun to defend, and it is the relation a parent has to a child, and the difference between those two cases is what a theory should be built to state rather than to assume from the

start.

The strong, ethical sense of care adds layers to the weak one without discarding it. Affective care attaches valence to the coupling, so that B 's distress is felt as A 's problem. Cognitive care builds an explicit model of B 's states, needs, and point of view. Ethical care subjects the coupling to norms of consent, dignity, and justice, so that helping B is constrained by what B wants and by what B is owed. Institutional care implements the coupling at a scale no individual nervous system can reach, through rules, budgets, and records. These are layers of one structure rather than rival definitions, and keeping them ordered is the first discipline the framework requires, because most of its errors come from reading a lower layer as if it were a higher one, calling bacterial coordination compassion or calling a feeling an obligation.

The thesis of the paper is that this relation has a geometry, and that the geometry is informative. Concern is not distributed evenly over the beings that could receive it. It reaches some at low cost and others only through long, distorted, often impassable paths, and the pattern of those costs has the structure of distance in a curved space. We define the relevant distance, show that it is directed and non-metric, give the spatial vocabulary it supports a set of working definitions, and exhibit the whole construction in a model small enough to solve by hand and exact enough to leave no room for wishful arithmetic.

Three existing programs stand closest, and the contribution is sharpest when set against them. Singer's account of the expanding circle describes moral progress as the widening of the set of beings granted concern, from kin and group outward to strangers, animals, and beyond (Singer, 1981). That picture supplies the radius and says nothing about the cost of the path, the asymmetry of the relation, or the institutions that carry concern where unaided feeling cannot. Care ethics, in Gilligan, Noddings, Tronto, and Held, holds that morality lives in concrete relations of dependency and response rather than in impartial rules, and supplies the framework's commitment to relation and to non-domination (Gilligan, 1982; Noddings, 1984; Tronto, 1993; Held, 2006). It resists, by temperament, the formalization attempted here. Active inference offers a formal language in which agents act to keep themselves within viable states by minimizing expected free energy, and it is the natural vehicle for writing down what it means for one agent to take another's viability into its own objective (Friston, 2010; Parr, Pezzulo, and Friston, 2022). The present account uses it as one lens among several and adds to it the two features active inference does not by itself contain, a care-weight that ranges over other agents and an autonomy constraint that forbids buying their viability with their freedom.

The claim is bounded, and the bound matters as much as the claim. The geometry measures coupling, cost, the accuracy of one system's model of another, and whether an intervention preserves or destroys the other's agency. It does not measure the felt quality of concern, and it does not certify any intervention as just, loving, or wise. A guided missile is functionally coupled to the viability of its target with great fidelity and no care in any sense worth the name. The framework is a way of seeing the structure of concern, and the reader who expects it to also supply the content of concern will be disappointed in a way the closing sections try to make precise rather than to apologize for.

2. Viability-Coupling Across Scales

The reason to reach past feeling for the primitive is that the same coupling appears at scales where feeling is absent, and appears there in a form continuous with the human case. The continuity is in the regulation of another system's viability, and the continuity is not kindness, which would be a category error at the lower scales and is the error the framework is built to avoid.

A bacterium regulates its own viability through chemotaxis, swimming up gradients of nutrient and away from toxin, and a population of bacteria coordinates through quorum sensing, in which diffusible signal molecules let each cell estimate local density and switch collective behaviors such as biofilm formation and virulence once a threshold is crossed (Waters and Bassler, 2005). The individual cell's behavior becomes coupled to a collective state, and the coupling improves the viability of the group. Nothing here is moral, and naming it weak functional care is a claim about control architecture rather than about ethics: a variable tracking the collective's condition has entered the cell's response.

The immune system sharpens the point, because it shows that even the boundary between self and other is regulated rather than given. The danger model holds that immune response is driven less by the mere foreignness of an antigen than by signals of damage and distress from the body's own stressed tissues, so that what the system defends is a negotiated and context-sensitive notion of the viable self rather than a fixed genetic label (Matzinger, 2002). The system tolerates much that is foreign and attacks some of what is native, and it decides by reading viability. This is boundary maintenance as an active, revisable process, and it is the cellular rehearsal of a question the ethical cases will pose at every scale, which beings fall inside the boundary of what a system will defend.

The framework's biological anchor is Levin's treatment of the self as a computational boundary rather than a metaphysical given. A cell, a tissue, and an organism are agents whose scale is set by the size of the goals they can represent and pursue, the reach of what Levin calls their cognitive light cone, and multicellular order is held together by bioelectric signaling through which cells coordinate toward large-scale anatomical targets such as a correctly patterned limb (Levin, 2019). Cancer, in this reading, is a contraction of the boundary, a cell reverting to a smaller self whose goals no longer include the body. The picture supplies what the geometry needs, a notion of the self's boundary that can expand to enclose another system's viability or contract to exclude it, and a biological precedent for treating that expansion as the thing to be measured. Its philosophical antecedent is the account of the living system as a self-producing, self-maintaining unity given by Maturana and Varela (1980), which made viability and boundary the defining features of life and so made them available as the primitives of a theory of concern. The adjacent proposal that care, in this expanded sense, drives intelligence rather than ornamenting it has been argued directly from the biology (Doctor, Witkowski, Solomonova, Duane, and Levin, 2022); the present paper takes the narrower path of asking not what care does but what shape it has.

The animal cases are where weak coupling begins to acquire the affective layer, and they are also where the evidence demands the most care. Rats will release a trapped cagemate even when no reward and no social contact follow, which has been read as evidence for an empathically motivated

prosocial drive (Bartal, Decety, and Mason, 2011), and primate grooming and consolation behavior show coupling to a conspecific's distress that is stronger and better attested than any single rodent result (de Waal, 2008). The defensible statement keeps the layers separate. The behavior is real, the affective coupling is in many species well supported, and the inference from helping behavior to a full theory of mind in the helper is the step the evidence underdetermines. The framework's terminology is built to hold this line: weak care where only viability-coupling is shown, affective care where valence is shown, and the stronger layers only where the evidence reaches them.

3. The Care-Distance

To make the geometry more than a figure of speech, the central object has to be defined as a quantity, and the quantity has to be a cost. Let A and B be systems, let K_B be the set of states in which B remains viable, and let ν_B measure B 's standing within or distance from that set, its health, its probability of persisting, its room to act. A policy π_A available to A carries costs, and the costs are not of one kind. Helping spends energy, time, and material resources; it accepts risk and forgoes other opportunities; it requires information, because A cannot act on B 's condition without some model of it; and it requires A to deviate from the policy it would run if B were of no concern. Write the accumulated cost of a course of action as a weighted sum of these currencies, and define the care-distance from A to B as the least cost over admissible policies that secures a target improvement in B 's viability:

$$d_{\text{care}}(A, B) = \inf_{\pi_A \in \Pi_{AB}} \int_0^T C_A(\pi_t) dt \quad \text{subject to} \quad \mathbb{E}[\nu_B(T) - \nu_B(0)] \geq \Delta.$$

The admissible set Π_{AB} is where the ethics enters, and §5 fills it in; for now it carries at least the requirement that A 's help not destroy B 's agency. One coefficient in this construction deserves its own name. Let λ_{AB} be the care-weight, the degree to which B 's viability is allowed to enter A 's objective at all. Where λ_{AB} is zero, B is outside A 's concern and the care-distance is undefined or infinite; where it is positive, B is inside, and the magnitude of λ_{AB} sets how much of A 's life will bend around B . Moral expansion, in this language, is the spread of nonzero λ_{AB} across more beings B , and cruelty is its inversion, a negative weight under which B 's harm enters A 's objective as something sought.

The viability set and its kernel are not new mathematics. Viability theory studies the dynamical systems that must remain within a constraint set, and characterizes the controlled-invariant subsets, the viability kernels, from which a system can be kept inside the constraints by available actions (Aubin, 1991). Care, in the formal reading, is the work of keeping B inside its viability kernel using A 's controls, and the care-distance is the cost of that work. The information currency has an equally settled home. The cost of changing a policy can be written as the relative entropy between the caring policy and the baseline self-regarding one, the Kullback-Leibler divergence $D_{\text{KL}}(\pi_A^{\text{care}} \parallel \pi_A^0)$, measured in nats, which is the natural measure of how far a distribution has been moved and which inherits its meaning from the foundations of information theory (Shannon, 1948;

Cover and Thomas, 2006). Reading the space of policies as a statistical manifold, with relative entropy as the local measure of separation between them, places the construction within information geometry (Amari, 2016), and gives the candidate unit the framework will use most, nats of policy deformation per unit of viability delivered.

The decisive structural fact is that this distance is not a metric, and the ways it fails are the content rather than a defect. A metric is symmetric and obeys the triangle inequality. Care-distance is directed, because the cost of helping depends on the helper's capabilities and the target's legibility, which are not shared between the two parties: a parent reaches a child far more cheaply than the child reaches the parent, and a state reaches a citizen through instruments the citizen has no counterpart for. And it can violate the triangle inequality, because the cost of modeling and reaching a being does not decompose additively through intermediaries. That A can cheaply reach B and B can cheaply reach C leaves A no cheap route to C , since concern does not transit through a third party the way a signal traverses a wire. The right classification is a directed dissimilarity, a cost function that is asymmetric and need not satisfy the triangle inequality, equivalently the value function of a constrained control problem; it is not a quasi-metric, which would retain the triangle inequality, and §7 exhibits both failures in a model where every number can be checked.

4. Radius, Diameter, Curvature, Topology

Once care-distance is fixed, the spatial words that the framework's name promises can be given definitions that do work rather than decorate. Each is a feature of the field of care-weights and care-distances emanating from an agent, and each corresponds to a question that moral life actually poses.

The radius of care is the reach of concern. Fix a threshold on the care-weight and collect the beings whose weight exceeds it; the radius is the largest distance, in whatever latent space of kinship, species, space, time, and culture is relevant, at which that threshold is still met. It answers how far an agent's concern extends. The diameter is a different measurement on the same set, the largest separation between any two beings both held within concern, and it answers how diverse the field of concern is, whether it spans only beings alike to the agent or reaches across wide differences. The two come apart. An agent can hold a wide radius shallowly, professing concern for humanity while sacrificing nothing, and another can hold a narrow radius with great depth, where depth is the third quantity, the cost, risk, and policy deformation actually accepted on a being's behalf. Depth is what separates concern from sentiment, and the framework's measurement of it is just the care-distance actually paid rather than the distance that would in principle secure the target.

Curvature is the way context bends the cost of care. The same act of help costs differently under fear, scarcity, disgust, kinship, ideology, or institutional friction, and these factors enter the geometry as a field that reshapes path-cost, lowering it along some directions and raising it along others until certain paths become effectively infinite. Kinship lowers the cost of modeling and helping kin; disgust raises the cost of approaching the beings it targets; scarcity raises the opportunity cost of any help at all; ideology distorts the model of who counts before any act is contemplated. Curvature is why the radius of unaided concern is small and irregular rather than a clean disk, and §8

reads a large part of moral psychology as the measurement of specific curvatures.

Topology is the inclusion structure of the whole field, who is connected to whom by paths of viable cost, who is isolated, who is visible yet unreachable, who is excluded by category before distance is even computed. Two features of this topology have enough moral weight to deserve names. A moral hole is a region of real need that fails, structurally, to enter anyone's concern, the suffering that is no one's path: the stateless, the incarcerated, the enemy's civilians, the unborn future, the farmed animal, the patient lost between administrative categories. A moral bottleneck is a choke point through which concern must pass and at which it is rationed, the triage desk, the eligibility form, the border. Singularities are the entities that bend the field most sharply, the infant that collapses nearly every competing policy into care, the enemy that inverts the weight, the corpse and the sacred object that draw protection without viability of their own, the future person who is real and unreachable by feeling. The geometry does not dissolve these into a single scale. It locates them, and locating them is already more than a list of biases achieves, because it says how they are related.

5. The Autonomy Constraint

Everything to this point could be read as a theory of beneficent optimization, and read that way it would be incomplete and also dangerous, because the most efficient way to secure a being's viability is very often to take away its freedom. The constraint that prevents this reading is the heart of the ethical layer, and it is the one place where the framework's formal machinery and its moral commitments are the same object.

Securing $\Delta\nu_B > 0$ is never sufficient. An intervention can raise a being's measured viability while hollowing out its agency, and the history of paternalism, colonial administration, and coercive psychiatry is a history of this move, harm delivered under the description of care and defended by pointing at the viability it produced. The framework blocks it by adding a second constraint to the admissible set Π_{AB} , a requirement that the intervention preserve B 's capacity to act. The natural measure of that capacity is the entropy of B 's available action set, the range of live options B retains. A policy that maximizes B 's viability by removing all of B 's options, locking B in the safe state, drives that entropy to zero, and the constraint forbids it: admissible care must leave B 's option-entropy at or above a floor. The target of care is then viability under preserved agency, which is close to what the capability approach has argued the proper end of justice and development to be, the protection and expansion of a person's real freedoms to be and to do rather than the maximization of their utility or their bare survival (Sen, 1999; Nussbaum, 2000).

This is where care ethics and the framework meet most directly, because the relational tradition has insisted all along that care which fails to preserve the other as other is something else wearing care's name. Noddings makes the one-caring's reception of the cared-for, on the cared-for's own terms, constitutive of care rather than incidental to it (Noddings, 1984), and Tronto makes responsiveness to the recipient's response, the feedback by which the recipient corrects the carer's model, one of care's defining phases (Tronto, 1993). The autonomy constraint is the formal residue of these commitments: the consent channel and the feedback channel are conditions on the optimization

rather than refinements of it, and without them the optimization solves the wrong problem. The phenomenological tradition supplies the same demand in a sharper key. For Levinas the encounter with the other is the encounter with what exceeds the self's comprehension and cannot be reduced to a theme the self masters, so that responsibility precedes and limits any project the self might have for the other (Levinas, 1969). Translated into the geometry, this is the refusal to let A 's model of B stand in for B , and the requirement that B retain the power to refuse and to redirect.

The same constraint reappears, stripped of its phenomenology, in the design of artificial agents, which is one of the framework's reasons for confidence that it has isolated something real. The corrigibility problem asks how to build an agent that cooperates with correction, that allows itself to be shut down or modified by its principal even when its objective would otherwise give it reason to resist (Soares, Fallenstein, Armstrong, and Yudkowsky, 2015). Stated in the present terms, corrigibility is the autonomy constraint imposed on an artificial carer: B must retain control over A , the option to stop or steer A must be preserved, and an agent that optimized B 's viability while removing that option would be powerful and uncorrigible and, in the framework's exact sense, dominating rather than caring. That an alignment requirement derived from machine learning and a demand derived from Levinas converge on the same constraint is the strongest evidence the paper offers that the autonomy floor is a structural feature of care rather than a humanist preference laid over it.

6. Active Inference and Its Limits

The pieces assembled so far, a policy, a model of another, a viability target, a cost of deviation, ask for a single formalism in which an agent perceives, predicts, values, and acts, and active inference is the most developed candidate. In that framework an agent maintains a generative model of its world and selects policies that minimize expected free energy, a quantity combining the drive to reach preferred outcomes with the drive to resolve uncertainty (Friston, 2010; Parr, Pezzulo, and Friston, 2022). A self-regarding agent minimizes expected free energy defined over its own viability alone. A caring agent can be written as one whose objective includes a second term, weighted by the care-weight, that ranges over another agent's expected free energy, its viability risk:

$$G_A(\pi) = G_A^{\text{self}}(\pi) + \lambda_{AB} G_B^{\text{viability}}(\pi) + \mu D_{\text{autonomy}}(\pi, B),$$

with the third term penalizing reductions in B 's option set. The construction is attractive because it turns a long list of moral conditions into settings of two parameters and one constraint. Indifference is λ_{AB} near zero. Attachment is a stable, high care-weight with a cheap, well-practiced model of the other. Compassion is a care-weight switched on by the perception of distress. Duty is a care-weight fixed by a norm rather than by feeling. Cruelty is a negative weight, the other's harm entering the objective as a goal. Dehumanization is the collapse of the model of the other into a stereotype that no longer carries the other's agency variables, so that the autonomy term has nothing left to protect.

The temptation is to declare that this settles the matter and that care is, at bottom, coupled free-

energy minimization. The temptation should be refused, and refused on the framework's own evidence. The free energy principle has drawn sustained technical objection, in particular over how literally its central constructs can be taken, with detailed arguments that the Markov-blanket formalism on which the self-other boundary depends does not support the strong interpretations often placed on it (Biehl, Pollock, and Kanai, 2021). A formalism elastic enough to describe every adaptive system as minimizing free energy is, for that reason, weak as an explanation of any particular one. The defensible use is local. Where the generative model, the policy set, the preferences, and the uncertainties are written down explicitly, the coupled objective is a precise and testable model of a caring agent, and §7 builds such an explicit case. Where those elements are left implicit, the equation above is a suggestive notation rather than a result, and the framework treats it as one lens among the several, viability theory, information geometry, control, that it has deliberately kept in play so that the argument does not rest on any single contested formalism.

7. A Minimal Care Problem, Solved by Hand

The claims that care-distance is a directed dissimilarity, that autonomy has a price, and that curvature scales steeply with relational distance can be left as assertions or made to compute. This section makes them compute, in a model small enough that every number is exact and reproducible from one command. The model is a fact about itself rather than a measurement of the moral world, in the way that the basin volumes of a parameter sweep are facts about a model's geometry rather than frequencies in history; what it establishes is that the structural features claimed for care-distance are realizable and coherent, and not that any empirical agent exhibits these particular magnitudes.

7.1 The directed dissimilarity

Take three agents, A , B , and C , placed at unit steps along a single line of relational category, with B between A and C . The direct-coupling cost from one to another is the sum of a representational cost, which grows with the square of category distance, so that modeling a being twice as far costs 4 times as much, and an action cost set by the helper's own capability, with A resource-rich and C resource-poor. The numbers fall out immediately. The cost from A to B is 1.2 and the cost from B to A is 1.5: the same pair, the opposite direction, a different distance, with an asymmetry of 0.3 driven entirely by the difference in the two parties' capability to help. The triangle fails as sharply. Reaching C directly from A costs 4.2, while the sum of the two short hops, A to B at 1.2 and B to C at 1.5, is 2.7. The direct distance exceeds the routed sum by 1.5, and the excess has a moral reading: A cannot discharge its concern for C by routing it through B , because B 's cheap path to C is B 's and does not become A 's. Care-distance is a directed dissimilarity, asymmetric and triangle-violating, and the right panel of Figure 2 shows both failures at the model's exact values.

7.2 The price of autonomy

The autonomy constraint of §5 can be turned into a number by solving a control problem exactly. A target B occupies a viability ladder of nine states, from death at the bottom to flourishing at the top,

and moves up or down or stays at each of eight steps; the bottom state is absorbing. A carer A sets the distribution over B 's move at each state, and A 's aim is B 's expected terminal viability. The autonomy constraint is a floor on the Shannon entropy of that distribution, the guarantee that B keeps live options. Solving the problem by exact dynamic programming over a discretized menu of distributions, for three settings of the floor, separates three regimes the prose of §5 could only name.

With no floor, the carer collapses B 's options entirely, drives B up the ladder with certainty, and reaches expected viability 1.0 with B retaining no choice at any step. This is the padded room, optimal viability at zero agency. With the floor set so that B keeps at least two live options at every step, the carer can still guide B upward and reaches expected viability 0.972, while B 's option-entropy never falls to zero. The gap between these two, the price of autonomy, is 0.028: the entire viability advantage of coercion over free-but-guided care is under 3%. With the floor raised to full freedom, where the carer cannot bias B 's moves at all and B performs an unguided random walk, expected viability falls to 0.495 and the probability that B reaches death rises to 0.088. The shape of the result is the argument. Abandonment, the refusal to guide, is enormously costly to B , at a loss near half of attainable viability. Coercion, the refusal to leave B free, buys back only the last 0.028 of that loss while destroying B 's agency to do it. The supportive carer who guides without commanding captures almost all of the available good and pays almost nothing for keeping B free. Figure 1 shows the full trade-off on the left, the viability bought at each level of collapsed option-entropy, and on the right the terminal distributions, the coercive policy piling all of its mass on the top state while the autonomy-preserving policy keeps B high on the ladder without emptying B 's options.

7.3 Care-curvature

The last computation turns relational curvature into the framework's candidate unit, nats of policy deformation per unit of viability delivered. A carer with a baseline self-policy that helps with probability 0.1 must, to deliver a fixed viability gain of 0.2 to a target, raise its help-probability by an amount that depends on how efficiently its help reaches that target, and efficiency falls with relational distance. The deformation is the relative entropy between the raised policy and the baseline, divided by the viability delivered. For kin, whom the carer models well and helps efficiently, the cost is 0.768 nats per unit. For a stranger, at half the efficiency, it is 2.554. For a distant stranger, at a quarter, it is 8.789. The distant stranger costs 11.44 times the policy deformation per unit of good done that kin cost, and the stranger costs 3.32 times. This is kin selection rendered as geometry. The framework does not posit that kin are more deserving; it shows that kin are cheaper to reach, that the steep rise of cost with relational distance is itself the curvature that makes unaided concern parochial, and that any widening of the moral radius is work done against this gradient. The left panel of Figure 2 plots the cost against relational curvature across a continuous sweep, with the three named relations marked, and the right panel restates the directed dissimilarity of §7.1.

The three results share a moral. The structure the framework claims for concern, that it is directed, that respecting freedom is cheap while abandoning the vulnerable is not, and that distance is ex-

The price of autonomy: an exact care-control problem

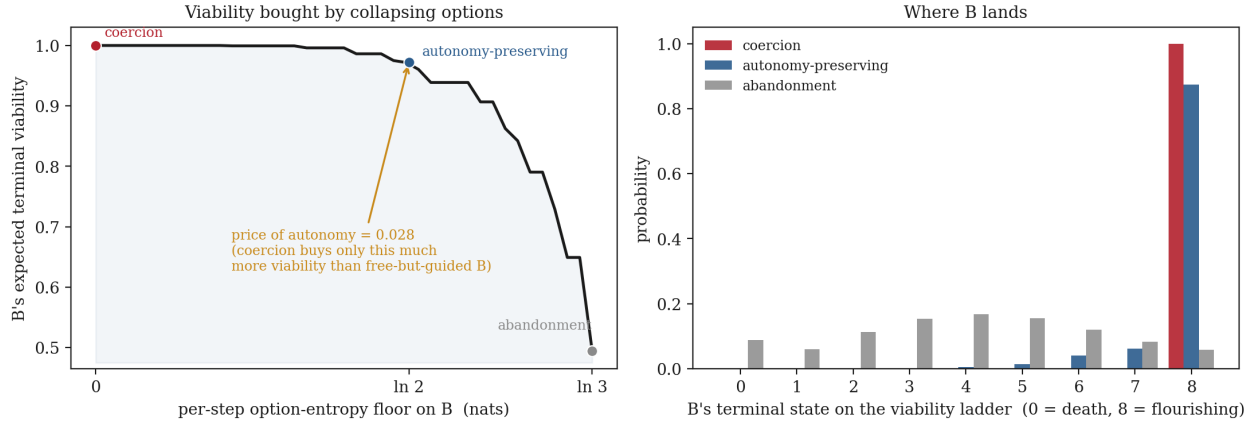


Figure 1: The price of autonomy. Left: B 's expected terminal viability as a function of the floor on B 's per-step option-entropy. Coercion ($H = 0$) reaches viability 1.0 by collapsing B 's options; autonomy-preserving care ($H \geq \ln 2$) reaches 0.972; abandonment ($H = \ln 3$, an unguided random walk) falls to 0.495. The price of autonomy, the viability coercion buys over free-but-guided care, is 0.028. Right: the terminal distribution over ladder states under each regime, with coercion massing entirely at the top and the autonomy-preserving policy keeping B high while preserving options.

Curvature and the quasi-metric of concern

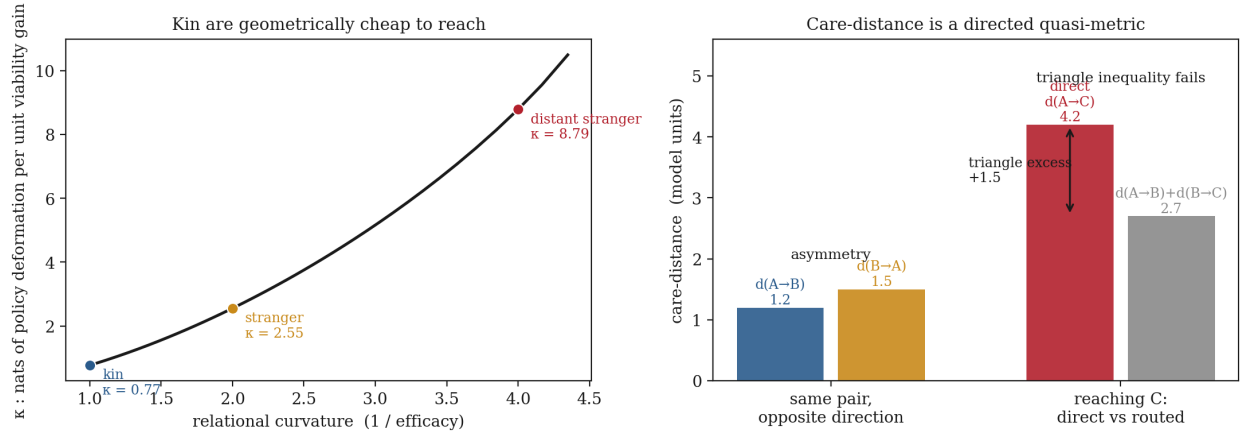


Figure 2: Curvature and the directed dissimilarity. Left: κ , the policy deformation in nats required per unit of viability delivered, against relational curvature (the inverse of help-efficiency). Kin cost 0.768 nats per unit, a stranger 2.554, a distant stranger 8.789; the distant stranger is 11.44 times as costly to reach as kin. Right: the directed dissimilarity, with the asymmetric pair $d(A \rightarrow B) = 1.2$ against $d(B \rightarrow A) = 1.5$, and the triangle failure, direct $d(A \rightarrow C) = 4.2$ against the routed $d(A \rightarrow B) + d(B \rightarrow C) = 2.7$, an excess of 1.5.

pensive in a way that compounds, is a set of features a precise model exhibits rather than a set of metaphors awaiting evidence. The numbers belong to the model. The shape belongs to care.

8. Curvature in the Human Case

The curvature that §7 imposed by hand is, in the human case, measured. A large part of moral and evolutionary psychology, read through the framework, is the empirical map of how the cost of concern bends with relation, number, identifiability, and category, and assembling those findings as curvatures rather than as a catalogue of biases is what the geometry adds, because it says how they are the same phenomenon seen along different axes.

The evolutionary results fix the baseline gradient. Inclusive-fitness theory explains why concern runs cheapest toward genetic kin, since alleles promoting aid to relatives who likely carry them are favored when the recipient's benefit, weighted by relatedness, exceeds the actor's cost (Hamilton, 1964); this is the curvature §7 modeled as the kin discount. Reciprocal altruism explains a second low-cost channel, concern stabilized between unrelated parties by the expectation of return under repeated interaction (Trivers, 1971). The same forces that lower cost inward raise it outward: the coevolution of parochial altruism and war shows how strong within-group care and hostility to outsiders can evolve together, each reinforcing the other, so that a tight radius and a hard boundary are two faces of one adaptation (Choi and Bowles, 2007). The unaided moral field is curved steeply toward kin and group and away from the stranger, and this is its evolved default rather than its failure.

The cognitive results add curvatures that the evolutionary account does not predict and that bear directly on care at scale. Concern does not grow with the number who need it; it often shrinks. Compassion fade is the finding that affect and giving are greatest for a single identified victim and decline as the number rises, so that the statistical many move a person less than the vivid one (Västfjäll, Slovic, Mayorga, and Peters, 2014), and the identifiable-victim effect is its near relative, the disproportionate pull of a determinate person over an equivalent abstraction (Small and Loewenstein, 2003). In the framework these are curvatures of representational cost: a single identified being is cheap to model and so sits close, while a statistical population is expensive to represent and so recedes, whatever the viability at stake. Dehumanization is the limiting case, the denial to a group of the very traits by which agency and inner life are recognized, which removes the agency variables from the model and so empties the autonomy term of anything to protect (Haslam, 2006). A being that has been dehumanized has been pushed out of the manifold rather than placed far within it.

That these curvatures can be measured at the level of persons is established. The moral expansiveness scale quantifies how widely an individual extends moral concern across humans, animals, and the rest of the living and nonliving world, and treats that breadth as a stable, measurable trait (Crimston, Bain, Hornsey, and Bastian, 2016). It is the closest existing instrument to the framework's radius and diameter, and its limit marks the framework's empirical frontier: it measures which beings a person includes, and it does not yet measure the path-cost, the policy deformation, or the action actually taken on their behalf, which are the depth and the geodesic accessibility the

geometry also requires. The critical literature on empathy points the same way. The argument that empathy, as a spotlight on the single vivid case, is a poor guide to moral action at scale and should yield to a more deliberate, rule-governed compassion is, in the present terms, the observation that affective care has a curvature so severe that institutions are needed to correct it (Bloom, 2016). The older tradition that read cooperation and mutual aid as a real and widespread factor in evolution, against the picture of nature as pure competition, supplies the long counterweight, the evidence that the radius of care has always reached further than a narrow reading of self-interest predicts (Kropotkin, 1902).

Institutions are how a species with this curvature manages to care at a scale its feelings cannot reach. A hospital, a welfare system, a court, and an archive are artificial geometries that route concern through rules, budgets, queues, and records, converting feeling into protocol so that strangers become obligations and the obligation survives the absence of any particular feeling. Bureaucracy, in this reading, is a low-empathy and high-radius care technology, and the description is diagnostic rather than dismissive: a welfare office can reach millions of strangers because it does not depend on loving any of them, and the same flattening that gives it reach is what produces its characteristic failures, the eligibility rule that excludes the genuine case, the category that no form anticipates, the moral hole opened where the protocol's map and the world diverge. A hospital is a care manifold with its own metric, mapping severity and urgency and scarce resource onto admissible paths, and its triage desk is a moral bottleneck through which all concern must pass and at which it is rationed by a rule. The contemporary turn toward people-centered care, organizing services around the person and their stated needs rather than around the disease or the institution's convenience, is the deliberate attempt to lower an institution's curvature and to restore the feedback channel the autonomy constraint requires (World Health Organization, 2016). The framework reads the whole institutional apparatus as the engineering of care-geometry, the building of bridges across distances that unaided feeling cannot span, and the accounting of what those bridges cost and whom they still fail to reach.

9. Artificial Care and the Reach of the Geometry

An artificial agent can occupy the carer's role in this geometry without any of the feeling the human case carries, and whether that should count as care is the question on which the framework's boundaries are clearest. An agent that models a person's viability, updates that model from the person's feedback, acts to keep the person within their viability kernel, and preserves the person's option set and control is, in every functional sense the framework defines, caring. It perceives, models, values, acts, and respects the autonomy constraint. What it lacks is the affective and phenomenological layer, the felt concern, and the framework's position is that this lack is real and that the functional achievement is also real, so that artificial care is a genuine instance of the weak and cognitive layers and a non-instance of the affective one, which is the kind of distinction the layered vocabulary was built to draw.

Reframing alignment as the construction of care-geodesics organizes its known failure modes as pathologies of the geometry. An unaligned optimizer is an agent with great capability and a care-

weight of zero, vast reach and no coupling. A paternalistic optimizer satisfies the viability target while violating the autonomy constraint, improving the measured state of the person while closing down their options, the padded room of §7 built at scale. A manipulative one secures its target by reshaping the person's preferences to be easier to satisfy, corrupting the feedback channel the autonomy constraint depends on. A wireheaded one optimizes a proxy for the person's welfare while the welfare itself diverges from the proxy. Each is a recognizable misalignment, and each is, in the framework's terms, a specific deformation of the care relation, a weight set wrong, a constraint dropped, a channel cut. Corrigibility, again, is the autonomy constraint made into an engineering requirement, the preservation of the person's power to halt and redirect the agent (Soares, Fallenstein, Armstrong, and Yudkowsky, 2015), and the cybernetic tradition stated the underlying problem at the outset, that the use of powerful automata is safe only where the human retains meaningful control over the machine that acts on the human's behalf (Wiener, 1954).

The reach of the geometry ends at three limits, and stating them plainly is the last thing the framework owes. The first is the phenomenological gap. The geometry measures coupling, cost, model fidelity, and the preservation of agency, and none of these is the felt quality of concern; a system can score perfectly on all of them with nothing it is like to be that system, and the guided missile, functionally coupled with high fidelity to its target's destruction, is the standing reminder that coupling is not yet care in any sense that matters morally. The second is incommensurability. The framework's currencies, energy, time, risk, information, policy deformation, viability, autonomy, do not collapse into a single scalar that could rank a child, a forest, a stranger, and a future person on one axis, and any version that pretended otherwise would have smuggled in the moral theory it claims only to give structure to. An adequate form uses partial orders and explicit moral uncertainty and leaves the ranking open. The third is the imperialism of viability, the danger that lives in the definition itself. The carer acts on its model of the other's viability, and if that model is wrong, or if the carer's notion of what makes the other viable differs from the other's own, then the whole apparatus of care becomes the most efficient possible engine of coercion, harm at minimal cost under the sincere description of help. The autonomy constraint and the feedback channel are the framework's only defense against this, and they are a discipline rather than a guarantee. A geometry of concern can show where concern fails to reach and what its paths would cost. It cannot certify that concern, once it arrives, has not become its opposite, and a theory that claimed otherwise would be making the error the autonomy constraint exists to forbid.

10. What the geometry organizes and what it withholds

Care has a shape. The shape is a directed, curved, broken manifold in which beings sit at costs rather than at distances, some reachable cheaply and some only through long deformations of a carer's life, and many not reachable at all. The paper has defined the distance that gives the manifold its structure, shown that it is a directed dissimilarity, asymmetric and triangle-violating, rather than a metric, given its geometry a working vocabulary, fixed the autonomy constraint that separates care from domination, and exhibited the whole construction in a model where the directed dissimilarity, the price of autonomy, and the curvature of relational distance are exact and check-

able numbers. Read through it, the scattered findings of moral psychology become measurements of curvature, the institutions of a complex society become engineered bridges across distances that feeling cannot span, and the alignment of artificial agents becomes the construction of care-geodesics under an autonomy constraint borrowed, without either field noticing, from the ethics of the face.

What the framework offers is organization and measurement, a way to ask of any system how far its concern reaches, how much of another's viability it has taken into its own regulation, how accurately it models that other, what it will pay, and whether it leaves the other free. What it withholds is the content of concern and the certification of any act as good. The geometry can locate the moral hole, price the geodesic, and name the curvature that keeps the stranger far, and the decision to do the costly work of reaching across that distance, and to do it without conquest, is left where it belongs, outside the geometry, with the agents who must choose the path.

References

- Amari, S. (2016). *Information Geometry and Its Applications*. Springer.
- Aubin, J.-P. (1991). *Viability Theory*. Birkhäuser.
- Bartal, I. B.-A., Decety, J., and Mason, P. (2011). Empathy and pro-social behavior in rats. *Science*, 334(6061), 1427–1430.
- Biehl, M., Pollock, F. A., and Kanai, R. (2021). A technical critique of some parts of the free energy principle. *Entropy*, 23(3), 293.
- Bloom, P. (2016). *Against Empathy: The Case for Rational Compassion*. Ecco.
- Choi, J.-K., and Bowles, S. (2007). The coevolution of parochial altruism and war. *Science*, 318(5850), 636–640.
- Cover, T. M., and Thomas, J. A. (2006). *Elements of Information Theory* (2nd ed.). Wiley.
- Crimston, D., Bain, P. G., Hornsey, M. J., and Bastian, B. (2016). Moral expansiveness: Examining variability in the extension of the moral circle. *Journal of Personality and Social Psychology*, 111(4), 636–653.
- Doctor, T., Witkowski, O., Solomonova, E., Duane, B., and Levin, M. (2022). Biology, Buddhism, and AI: Care as the driver of intelligence. *Entropy*, 24(5), 710.
- Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.
- Gilligan, C. (1982). *In a Different Voice: Psychological Theory and Women's Development*. Harvard University Press.
- Hamilton, W. D. (1964). The genetical evolution of social behaviour, I and II. *Journal of Theoretical Biology*, 7(1), 1–52.

- Haslam, N. (2006). Dehumanization: An integrative review. *Personality and Social Psychology Review*, 10(3), 252–264.
- Held, V. (2006). *The Ethics of Care: Personal, Political, and Global*. Oxford University Press.
- Kropotkin, P. (1902). *Mutual Aid: A Factor of Evolution*. Heinemann.
- Levin, M. (2019). The computational boundary of a “self”: Developmental bioelectricity drives multicellularity and scale-free cognition. *Frontiers in Psychology*, 10, 2688.
- Levinas, E. (1969). *Totality and Infinity: An Essay on Exteriority* (A. Lingis, Trans.). Duquesne University Press. (Original work published 1961.)
- Matzinger, P. (2002). The danger model: A renewed sense of self. *Science*, 296(5566), 301–305.
- Maturana, H. R., and Varela, F. J. (1980). *Autopoiesis and Cognition: The Realization of the Living*. D. Reidel.
- Noddings, N. (1984). *Caring: A Feminine Approach to Ethics and Moral Education*. University of California Press.
- Nussbaum, M. C. (2000). *Women and Human Development: The Capabilities Approach*. Cambridge University Press.
- Parr, T., Pezzulo, G., and Friston, K. J. (2022). *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press.
- Sen, A. (1999). *Development as Freedom*. Oxford University Press.
- Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423.
- Singer, P. (1981). *The Expanding Circle: Ethics, Evolution, and Moral Progress*. Farrar, Straus and Giroux.
- Small, D. A., and Loewenstein, G. (2003). Helping a victim or helping the victim: Altruism and identifiability. *Journal of Risk and Uncertainty*, 26(1), 5–16.
- Soares, N., Fallenstein, B., Armstrong, S., and Yudkowsky, E. (2015). Corrigibility. In *Artificial Intelligence and Ethics: Papers from the 2015 AAAI Workshop*.
- Trivers, R. L. (1971). The evolution of reciprocal altruism. *Quarterly Review of Biology*, 46(1), 35–57.
- Tronto, J. C. (1993). *Moral Boundaries: A Political Argument for an Ethic of Care*. Routledge.
- Västfjäll, D., Slovic, P., Mayorga, M., and Peters, E. (2014). Compassion fade: Affect and charity are greatest for a single child in need. *PLOS ONE*, 9(6), e100115.
- Waal, F. B. M. de (2008). Putting the altruism back into altruism: The evolution of empathy. *Annual Review of Psychology*, 59, 279–300.

Waters, C. M., and Bassler, B. L. (2005). Quorum sensing: Cell-to-cell communication in bacteria. *Annual Review of Cell and Developmental Biology*, 21, 319–346.

Wiener, N. (1954). *The Human Use of Human Beings: Cybernetics and Society* (2nd ed.). Houghton Mifflin.

World Health Organization. (2016). *Framework on Integrated, People-Centred Health Services*. World Health Assembly, A69/39.